



1) Calculate the number of moles of each of the following substances. Give your answers to 3 significant figures.

- a) 90.0 g of H_2O $\frac{90.0}{18} = 5.00$
- b) 20.0 g of C_4H_{10} $\frac{20.0}{58} = 0.345$
- c) 685 g of NH_3 $\frac{685}{17} = 40.3$
- d) 102 tons of O_2 $\frac{102000000}{32} = 3190000 \text{ (} 3.19 \times 10^6 \text{)}$
- e) 2.00 kg of Al_2O_3 $\frac{2000}{102} = 19.6$
- f) 20.6 mg of Au $\frac{0.0206}{197} = 0.000105 \text{ (} 1.05 \times 10^{-4} \text{)}$

2) Calculate the mass of each of the following substances. Give your answers to 3 significant figures.

- a) 4.00 moles of N_2 $4.00 \times 28 = 112 \text{ g}$
- b) 0.100 moles of HNO_3 $0.100 \times 63 = 6.30 \text{ g}$
- c) 0.0200 moles of K_2O $0.0200 \times 94 = 1.88 \text{ g}$
- d) 2.50 moles of PH_3 $2.50 \times 34 = 85.0 \text{ g}$
- e) 0.400 moles of $\text{C}_2\text{H}_5\text{OH}$ $0.400 \times 46 = 18.4 \text{ g}$
- f) 10.0 moles of Ca(OH)_2 $10.0 \times 74 = 740 \text{ g}$

3) 0.0200 moles of a compound is found to have a mass of 1.64 g. Find the formula mass of the compound.

$$\text{Formula mass} = \frac{1.64}{0.0200} = 82.0$$

Area	Strength	To develop	Area	Strength	To develop	Area	Strength	To develop
Done with care and thoroughness			Can find moles from mass			Can convert units		
Shows suitable working			Can find mass from moles			Can find M_r from mass and moles		
Can work out M_r			Can use sig figs			Gives units		